

Use Percent to Solve Problems

Lesson 4-5

Name: _____

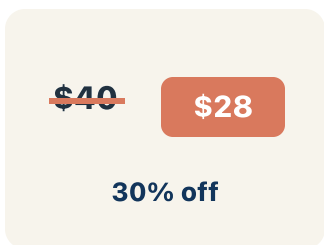
Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

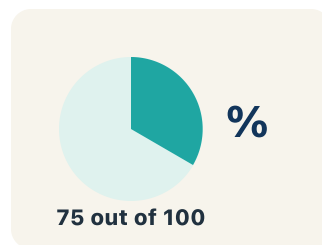
Picture first, then the word, then a plain-language meaning. Say each word out loud.



20% off \$50 → discount is \$10

Discount

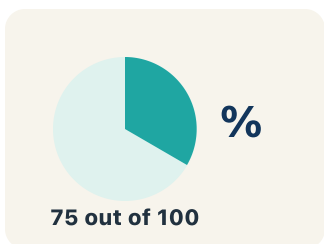
Write the definition:



A \$30 item with 10% markup sells for \$33

Markup

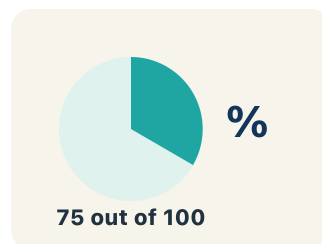
Write the definition:



6% tax on \$20 adds \$1.20

Tax

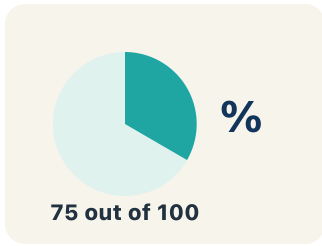
Write the definition:



15% tip on \$40 = \$6

Tip

Write the definition:



50% means 50 out of 100

Percent

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can solve problems involving discounts, markups, tax, and tips using percents.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Discount

Markup

Tax

Tip

Percent

1

An amount taken off the original price is a _____.

2

An amount added to a store's cost to set a higher selling price is a _____.

3

An extra percent added to a purchase that goes to the government is _____.

4

Extra money added for good service, often a percent of the bill, is a _____.

5

A number out of 100 is a _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: A prize costs 120 tickets and is 25% off. What is the sale price?

- A. 90 tickets
- B. 95 tickets
- C. 100 tickets
- D. 30 tickets

Step 1 25% of 120 = $0.25 \times 120 = 30$.

Step 2 Sale price: $120 - 30 = 90$ tickets.

 **Answer:** A. 90 tickets

 We do — together

Solve this one with your class using the same steps.

Problem: A meal costs \$20. You leave a 15% tip. How much is the tip?

- A. \$3.00
- B. \$1.50
- C. \$5.00
- D. \$0.15

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: A shirt costs \$40 and the sales tax is 5%. What is the total cost including tax?

- A. \$42.00
- B. \$45.00
- C. \$40.05
- D. \$38.00

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Store 2 — Tech Stop: A \$25 phone case has 8% sales tax added. What is the total cost?

- A. \$27
- B. \$33
- C. \$25.08
- D. \$23

Show your work:

2. Store 3 — Food Court: Your lunch bill is \$40. You leave a 15% tip. How much is the tip?

- A. \$6
- B. \$4
- C. \$15
- D. \$46

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A store has a 20% off sale, then takes an additional 10% off the sale price. Is this the same as 30% off the original? Use \$100 as the original price. Show your work for both scenarios and explain the difference.

Sentence starter: Taking 20% off then 10% off gives a final price of \$____, but 30% off the original gives \$____. The difference is \$____ because ____.

Show your work:

Reflect — Exit Ticket

A jacket costs \$80 and is 35% off. Sales tax is 5%. What is the final price?

- A. \$54.60
- B. \$52.00
- C. \$56.00
- D. \$49.40

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Discount
2. **Notes 2:** Markup
3. **Notes 3:** Tax
4. **Notes 4:** Tip
5. **Notes 5:** Percent
6. **We Try (notes):** A. $\$3.00 - 15\% \text{ of } \$20 = 0.15 \times 20 = \$3.00$.
7. **You Try (notes):** A. $\$42.00 - \text{Tax: } 0.05 \times \$40 = \$2.00$. *Total: $\$40 + \$2 = \$42.00$.*
8. **Try It 1:** A. $\$27 - \text{Tax: } 0.08 \times 25 = 2$. *Total: $25 + 2 = \$27$.*
9. **Try It 2:** A. $\$6 - \text{Tip: } 0.15 \times 40 = \6 .
10. **Exit Ticket:** A. $\$54.60 - \text{Discount: } 0.35 \times \$80 = \$28$. *Sale price: $\$80 - \$28 = \$52$. Tax: $0.05 \times \$52 = \2.60 . Final: $\$52 + \$2.60 = \$54.60$.*